

# LAB EXPERIMENT RECORD

## Set 7: Customer Demographics Analysis

|                           |                                           |
|---------------------------|-------------------------------------------|
| Department                | CSE-Ds                                    |
| Course Code / Course Name | DSA0601 / Data Handling and Visualization |
| Experiment                | Data Visualization using R                |
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| Day                       | Saturday                                  |

### Aim

To analyze and visualize customer demographic data using a bar chart, pie chart, and table, and to create an interactive dashboard for exploring customer demographics.

### Dataset: Customer Demographics

| Customer ID | Age | Gender | Income (in \$) |
|-------------|-----|--------|----------------|
| 1           | 28  | Female | 50000          |
| 2           | 35  | Male   | 60000          |
| 3           | 42  | Female | 75000          |

### PROGRAM 1: Bar Chart of Customer Ages

Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

### R Program

R

```
customer_id <- c(1, 2, 3)
```

```
age <- c(28, 35, 42)
```

```
barplot(age,
```

```
  names.arg = customer_id,
```

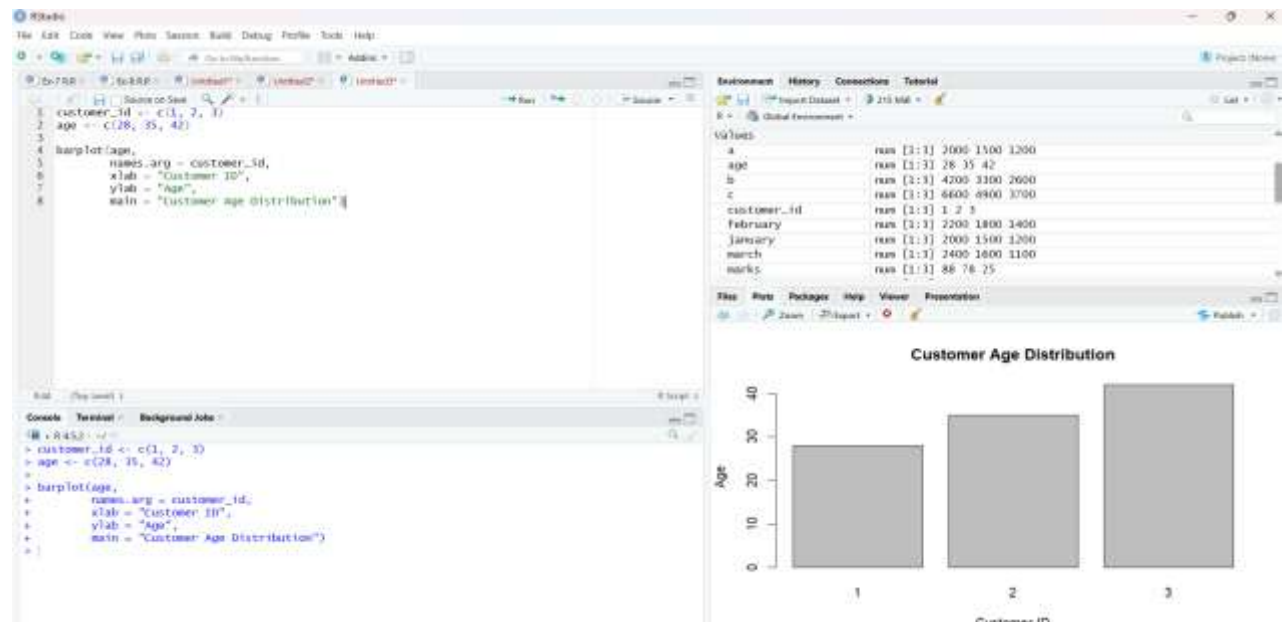
```
  col = "skyblue",
```

```
  xlab = "Customer ID",
```

```
  ylab = "Age",
```

```
  main = "Customer Age Distribution")
```

### Output 1: Customer Age Distribution Bar Chart



### Observation

The bar chart shows the ages of the three customers. Customer 1 is 28 years old, Customer 2 is 35 years old, and Customer 3 is 42 years old. Customer 3 has the highest age.

### PROGRAM 2: Pie Chart of Customer Gender

Generate a pie chart to represent the distribution of customers by gender.

#### R Program

R

```
gender <- c("Female", "Male")
```

```
count <- c(2, 1)

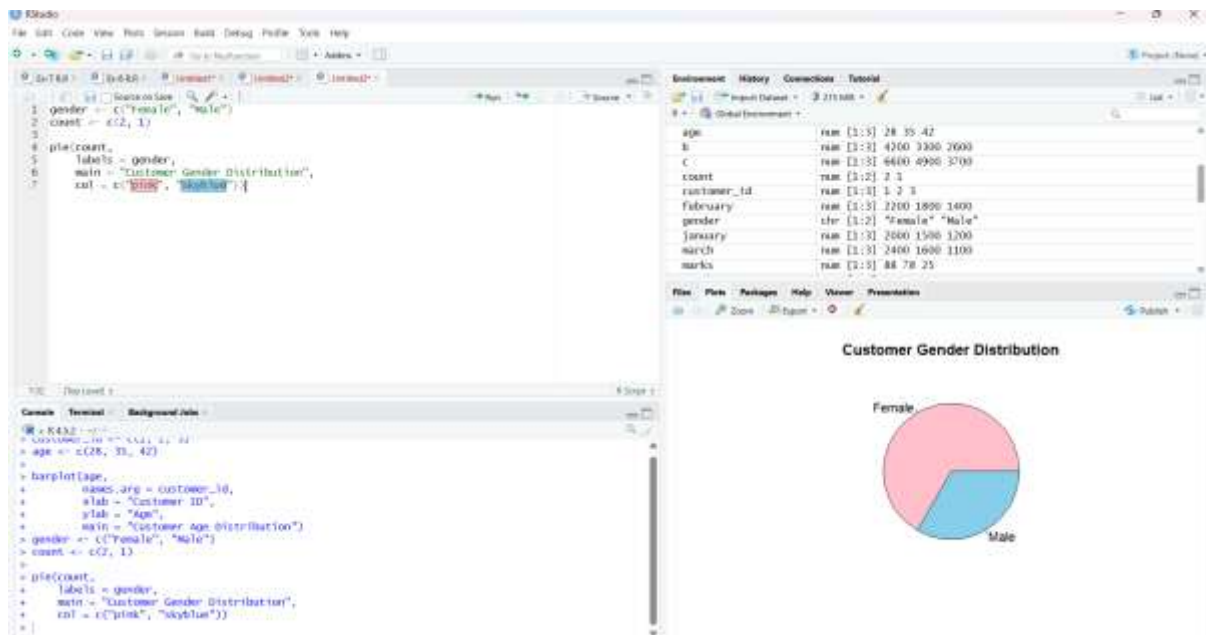
pie(count,

    labels = gender,

    main = "Customer Gender Distribution",

    col = c("pink", "skyblue"))
```

## Output 2: Customer Gender Distribution Pie Chart



## Observation

The pie chart represents the gender distribution of the customers. There are two female customers and one male customer. Therefore, females form the larger portion of the dataset.

## PROGRAM 3: Customer Demographic Table

Build a table to show the demographic information for each customer. Label the table elements.

## R Program

R

```
customer_id <- c(1, 2, 3)

age <- c(28, 35, 42)

gender <- c("Female", "Male", "Female")

income <- c(50000, 60000, 75000)
```

```
customer_data <- data.frame(  
  Customer_ID = customer_id,  
  Age = age,  
  Gender = gender,  
  Income = income  
)  
print(customer_data)
```

**Output 3: Customer Demographic Table**

The screenshot shows the RStudio interface. The source editor contains the following R code:

```
1 customer_id <- c(1, 2, 3)
2 age <- c(28, 35, 42)
3 gender <- c("Female", "Male", "Female")
4 income <- c(50000, 60000, 75000)
5
6 customer_data <- data.frame(
7   Customer_ID = customer_id,
8   Age = age,
9   Gender = gender,
10  Income = income
11 )
12
13 print(customer_data)
```

The console output shows the execution of the code and the resulting data frame:

```
> gender <- c("Female", "Male", "Female")
> income <- c(50000, 60000, 75000)
>
> customer_data <- data.frame(
+   Customer_ID = customer_id,
+   Age = age,
+   Gender = gender,
+   Income = income
+ )
>
> print(customer_data)
  Customer_ID Age Gender Income
1           1  28 Female 50000
2           2  35   Male 60000
3           3  42 Female 75000
> |
```

## Observation

The table displays the complete demographic information of each customer, including Customer ID, age, gender, and income.

## PROGRAM 4: Interactive Customer Demographics Dashboard

Develop an interactive dashboard combining the bar chart, pie chart, and demographic table for customer exploration.

## R Program

First install Shiny **only once**:

R

```
install.packages("shiny")
```

Then use this complete code:

R

```
library(shiny)
```

```
# Customer Data
```

```
customer_id <- c(1, 2, 3)
```

```
age <- c(28, 35, 42)
```

```
gender <- c("Female", "Male", "Female")
```

```
income <- c(50000, 60000, 75000)
```

```
customer_data <- data.frame(
```

```
  Customer_ID = customer_id,
```

```
  Age = age,
```

```
  Gender = gender,
```

```
  Income = income
```

```
)
```

```
# User Interface
```

```
ui <- fluidPage(
```

```
  titlePanel("Customer Demographics Analysis Dashboard"),
```

```
  sidebarLayout(
```

```
    sidebarPanel(
```

```
      selectInput(
```

```
        "customer",
```

```
        "Select Customer:",
```

```
        choices = c("All", "1", "2", "3"),
```

```
        selected = "All"
```

```
)
```

```

),
mainPanel(

  fluidRow(
    column(
      6,
      h3("Customer Age Distribution"),
      plotOutput("ageChart", height = "350px")
    ),
    column(
      6,
      h3("Gender Distribution"),
      plotOutput("genderChart", height = "350px")
    )
  ),
  hr(),
  h3("Customer Demographic Information"),
  tableOutput("customerTable")
)))

server <- function(input, output) {
  output$ageChart <- renderPlot({
    if (input$customer == "All") {
      data <- customer_data
    } else {
      data <- customer_data[
        customer_data$Customer_ID == as.numeric(input$customer),
        ]
    }
  })
}

```

```

}

barplot(

  data$Age,

  names.arg = data$Customer_ID,

  col = "skyblue",

  ylim = c(0, 50),

  xlab = "Customer ID",

  ylab = "Age",

  main = "Customer Age Distribution"

)

})

output$genderChart <- renderPlot({

if (input$customer == "All") {

gender_count <- table(customer_data$Gender)

} else {

  data <- customer_data[

    customer_data$Customer_ID == as.numeric(input$customer),

  ]

  gender_count <- table(data$Gender)

}

pie(

  gender_count,

  main = "Customer Gender Distribution",

  col = c("pink", "skyblue")

)

})

```

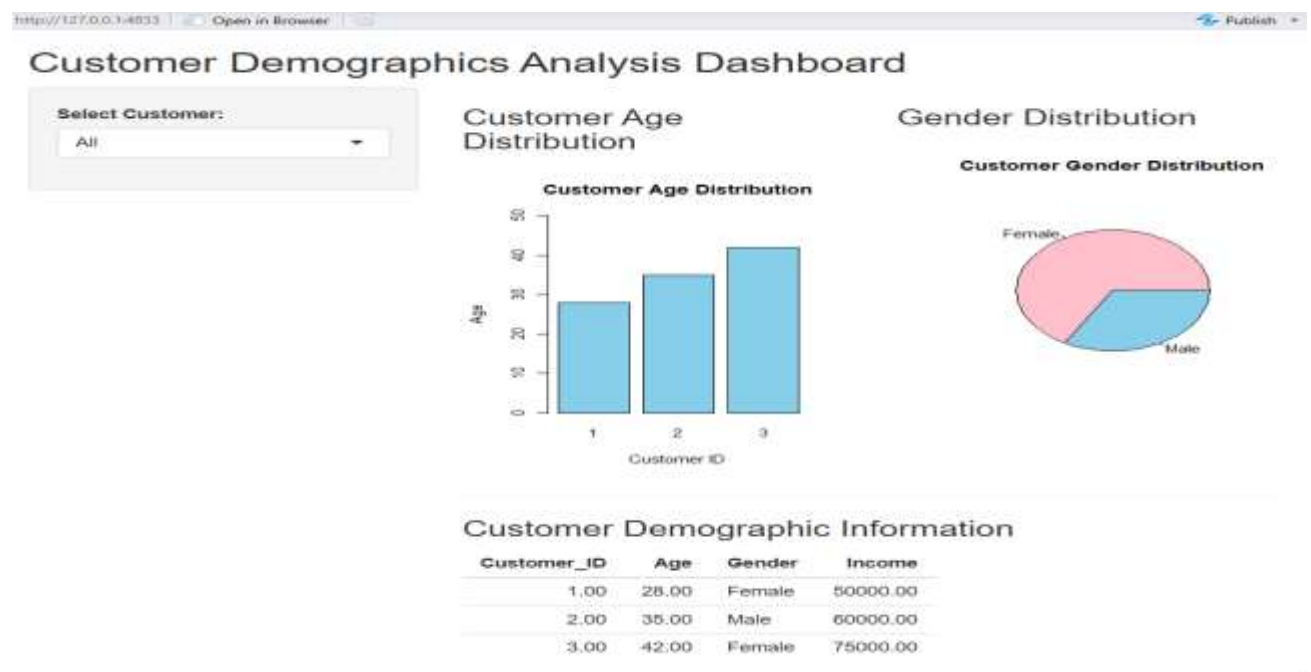
```

output$customerTable <- renderTable({
  if (input$customer == "All") {
    customer_data
  } else {
    customer_data[
      customer_data$Customer_ID == as.numeric(input$customer),
    ]
  }
})
}

shinyApp(ui = ui, server = server)

```

#### OUTPUT 4:



The **Select Customer** dropdown makes it interactive. If you select Customer 1, 2, or 3, the chart and table update.